DEEP LEARNING LABORATORY

Experiment 4 - CNN and Learned Filter Visualization

A Laboratory Experiment Report

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Dataset
CIFAR-10

Prepared from the captured experiment notebook outputs

Experiment Overview

Experiment: Experiment 04

Dataset: CIFAR-10

CNN and Learned Filter Visualization

This is a standalone notebook for this lab section. Run the setup and dataset cells first, then run the experiment cell and its output check. Use a Colab GPU runtime for the image and generative-model experiments when available.

Step 1 - Install the Additional Colab Packages

Wait until installation completes before the next cell starts.

Code

```
%pip -q install -U datasets "transformers<5" sentencepiece accelerate torchmetrics torch-fidelity ucimlrepo tabulate
```

Step 2 - Verify the Runtime

This cell records the Python, TensorFlow, and PyTorch versions and enables safe TensorFlow GPU memory growth.

Code

```
from pathlib import Path
import platform
import tensorflow as tf
import torch

project_root = Path("/content/Deeplearning")
project_root.mkdir(parents=True, exist_ok=True)
for gpu in tf.config.list_physical_devices("GPU"):
    try:
        tf.config.experimental.set_memory_growth(gpu, True)
    except RuntimeError:
        pass
print(f"Python: {platform.python_version()}")
print(f"TensorFlow: {tf.__version__}")
print(f"PyTorch: {torch.__version__}")
print(f"CUDA available: {torch.cuda.is_available()}")
print(f"Project root: {project_root}")
```

```
Python: 3.13.15
TensorFlow: 2.20.0
PyTorch: 2.11.0+cu128
CUDA available: True
Project root: /content/Deeplearning
```

Step 3 - Import Every Required Dataset

This cell creates visible files under `/content/Deeplearning/datasets` and prints a manifest that can be shown to the staff.

- MNIST, Iris, CIFAR-10, Dogs vs. Cats, Airline Passengers, and Tiny Shakespeare are downloaded as complete or balanced lab-ready files.

- MS COCO, CelebA, and LSUN Bedrooms are the genuine named datasets, streamed as practical subsets for Colab.

- No synthetic replacement dataset is used.

Code

```
from pathlib import Path
import json
import shutil
import numpy as np
```

```
import pandas as pd
import tensorflow as tf
from PIL import Image
from datasets import load_dataset
from sklearn.datasets import load_iris


root = Path("/content/Deeplearning") if Path("/content").exists() else Path(__file__).resolve().parent
dataset_dir = root / "datasets"
output_dir = root / "outputs"
dataset_dir.mkdir(parents=True, exist_ok=True)
output_dir.mkdir(parents=True, exist_ok=True)


def balanced_indices(labels, per_class, seed):
    labels = np.asarray(labels).reshape(-1)
    indices = np.concatenate([np.flatnonzero(labels == class_index)[:per_class] for class_index in np.unique(labels)])
    rng = np.random.default_rng(seed)
    rng.shuffle(indices)
    return indices


def save_image_stream(dataset_name, split, destination, count, size, config=None):
    destination.mkdir(parents=True, exist_ok=True)
    existing = sorted(destination.glob("*.jpg"))
    if len(existing) >= count:
        return len(existing)
    arguments = {
        "path": dataset_name,
        "split": split,
        "streaming": True,
    }
    if config is not None:
        arguments["name"] = config
    stream = load_dataset(**arguments)
    saved = 0
    for row in stream.take(count):
        image = row["image"].convert("RGB")
        image = image.resize((size, size), Image.Resampling.LANCZOS)
        image.save(destination / f"image_{saved:04d}.jpg", quality=92)
        saved += 1
    if saved < count:
        raise RuntimeError(f"Only {saved} images were available from {dataset_name}")
    return saved


(mnist_x_train, mnist_y_train), (mnist_x_test, mnist_y_test) = tf.keras.datasets.mnist.load_data()
mnist_train_indices = balanced_indices(mnist_y_train, 1000, 42)
mnist_test_indices = balanced_indices(mnist_y_test, 200, 42)
np.savez_compressed(
    dataset_dir / "mnist_lab.npz",
    x_train=mnist_x_train[mnist_train_indices],
    y_train=mnist_y_train[mnist_train_indices],
    x_test=mnist_x_test[mnist_test_indices],
    y_test=mnist_y_test[mnist_test_indices],
)

try:
    from ucimlrepo import fetch_ucirepo

    iris_repository = fetch_ucirepo(id=53)
    iris_features = iris_repository.data.features.copy()
    iris_features.columns = ["sepal_length", "sepal_width", "petal_length", "petal_width"]
    iris_target = iris_repository.data.targets.iloc[:, 0].astype(str)
    iris_target = iris_target.str.replace("Iris-", "", regex=False).str.replace("Iris ", "", regex=False).str.lower()
    iris = iris_features.copy()
    iris["species"] = iris_target.to_numpy()
    iris_source = "UCI Machine Learning Repository dataset 53"
except Exception:
    iris_bundle = load_iris(as_frame=True)
    iris = iris_bundle.data.copy()
    iris.columns = ["sepal_length", "sepal_width", "petal_length", "petal_width"]
    iris["species"] = pd.Series(iris_bundle.target).map(dict(enumerate(iris_bundle.target_names)))
    iris_source = "scikit-learn packaged Iris fallback"
iris.to_csv(dataset_dir / "iris.csv", index=False)

(cifar_x_train, cifar_y_train), (cifar_x_test, cifar_y_test) = tf.keras.datasets.cifar10.load_data()
cifar_train_indices = balanced_indices(cifar_y_train, 800, 42)
cifar_test_indices = balanced_indices(cifar_y_test, 200, 42)
np.savez_compressed(
    dataset_dir / "cifar10_lab.npz",
    x_train=cifar_x_train[cifar_train_indices],
    y_train=cifar_y_train[cifar_train_indices].reshape(-1),
    x_test=cifar_x_test[cifar_test_indices],
    y_test=cifar_y_test[cifar_test_indices].reshape(-1),
)
```

```
def build_cats_and_dogs_filtered(destination, image_size=160, seed=42):
    """Rebuild the classic 'cats_and_dogs_filtered' train/validation layout.

    The original tutorial archive at
    https://storage.googleapis.com/mledu-datasets/cats_and_dogs_filtered.zip
    now returns HTTP 403 (Google retired the mledu-datasets bucket), so the
    same 3,000-image split is reconstructed by streaming labelled images
    from the public "microsoft/cats_vs_dogs" dataset on the Hugging Face Hub.
    """
    targets = {
        destination / "train" / "cats": (0, 1000),
        destination / "train" / "dogs": (1, 1000),
        destination / "validation" / "cats": (0, 500),
        destination / "validation" / "dogs": (1, 500),
    }
    already_done = all(
        len(list(folder.glob("*.jpg"))) >= count for folder, (_, count) in targets.items()
    )
    if already_done:
        return destination
    for folder in targets:
        folder.mkdir(parents=True, exist_ok=True)
    counts = {folder: 0 for folder in targets}
    stream = load_dataset("microsoft/cats_vs_dogs", split="train", streaming=True)
    stream = stream.shuffle(seed=seed, buffer_size=2000)
    for row in stream:
        if all(counts[folder] >= count for folder, (_, count) in targets.items()):
            break
        label = row["labels"]
        for folder, (target_label, target_count) in targets.items():
            if label != target_label or counts[folder] >= target_count:
                continue
            try:
                image = row["image"].convert("RGB")
            except Exception:
                break
            image = image.resize((image_size, image_size), Image.Resampling.LANCZOS)
            image.save(folder / f"image_{counts[folder]:04d}.jpg", quality=92)
            counts[folder] += 1
            break
    for folder, (_, count) in targets.items():
        if counts[folder] < count:
            raise RuntimeError(f"Only {counts[folder]} images were available for {folder}")
    return destination


dogs_cats_dir = dataset_dir / "cats_and_dogs_filtered"
build_cats_and_dogs_filtered(dogs_cats_dir)
if not (dogs_cats_dir / "train").exists():
    raise FileNotFoundError("Dogs vs. Cats extraction folder was not created")

airline_url = "https://raw.githubusercontent.com/jbrownlee/Datasets/refs/heads/master/monthly-airline-passengers.csv"
airline = pd.read_csv(airline_url)
airline.to_csv(dataset_dir / "monthly_airline_passengers.csv", index=False)

shakespeare_cache = tf.keras.utils.get_file(
    fname="shakespeare.txt",
    origin="https://storage.googleapis.com/download.tensorflow.org/data/shakespeare.txt",
    cache_dir=str(root),
    cache_subdir="downloads",
)
shakespeare_path = dataset_dir / "shakespeare.txt"
shutil.copyfile(shakespeare_cache, shakespeare_path)

coco_dir = dataset_dir / "ms_coco_subset"
coco_dir.mkdir(parents=True, exist_ok=True)
coco_caption_path = coco_dir / "captions.csv"
if not coco_caption_path.exists() or len(list(coco_dir.glob("coco_*.jpg"))) < 5:
    coco_stream = load_dataset("mm-eval/MS-COCO-Captions", split="val", streaming=True)
    coco_rows = []
    for index, row in enumerate(coco_stream.take(5)):
        image_path = coco_dir / f"coco_{index:03d}.jpg"
        media = row["media"]
        image = media[0] if isinstance(media, list) else media
        image.convert("RGB").save(image_path, quality=92)
        coco_rows.append(
            {
                "file": image_path.name,
                "captions_json": json.dumps(list(row["answer"]), ensure_ascii=False),
            }
        )
    pd.DataFrame(coco_rows).to_csv(coco_caption_path, index=False)

celeba_dir = dataset_dir / "celeba_subset"
celeba_count = save_image_stream("flwrlabs/celeba", "train", celeba_dir, 384, 64, "img_align+identity+attr")
```

```
lsun_dir = dataset_dir / "lsun_bedrooms_subset"
lsun_count = save_image_stream("pcuenq/lsun-bedrooms", "train", lsun_dir, 512, 64)

manifest = pd.DataFrame(
    [
        {"dataset": "MNIST", "path": "mnist_lab.npz", "records": 12000, "experiments": "1", "source": "TensorFlow Keras
            MNIST", "scope": "Balanced lab subset"},
        {"dataset": "Iris", "path": "iris.csv", "records": len(iris), "experiments": "2", "source": iris_source,
            "scope": "Complete dataset"},
        {"dataset": "CIFAR-10", "path": "cifar10_lab.npz", "records": 10000, "experiments": "3, 4, Mini Project",
            "source": "TensorFlow Keras CIFAR-10", "scope": "Balanced lab subset"},
        {"dataset": "Dogs vs. Cats", "path": str(dogs_cats_dir.relative_to(dataset_dir)), "records": 3000,
            "experiments": "5", "source": "microsoft/cats_vs_dogs public mirror (Hugging Face)", "scope": "Filtered
            dataset (rebuilt 1000/1000 train, 500/500 validation)"},
        {"dataset": "Airline Passengers", "path": "monthly_airline_passengers.csv", "records": len(airline),
            "experiments": "6", "source": airline_url, "scope": "Complete dataset"},
        {"dataset": "MS COCO", "path": "ms_coco_subset", "records": 5, "experiments": "7", "source":
            "mm-eval/MS-COCO-Captions public parquet mirror", "scope": "Real-image demonstration subset"},
        {"dataset": "CelebA", "path": "celeba_subset", "records": celeba_count, "experiments": "8", "source":
            "flwrlabs/celeba public mirror", "scope": "Streamed training subset"},
        {"dataset": "Shakespeare Corpus", "path": "shakespeare.txt", "records":
            len(shakespeare_path.read_text(encoding="utf-8")), "experiments": "9", "source": "TensorFlow Tiny
            Shakespeare", "scope": "Complete corpus"},
        {"dataset": "LSUN Bedrooms", "path": "lsun_bedrooms_subset", "records": lsun_count, "experiments": "10",
            "source": "pcuenq/lsun-bedrooms public mirror", "scope": "Streamed training subset"},
    ]
)
manifest.to_csv(dataset_dir / "dataset_manifest.csv", index=False)

print(f"Deeplearning root: {root}")
print(manifest.to_string(index=False))
print("Dataset evidence:")
for path in sorted(dataset_dir.iterdir()):
    if path.is_dir():
        file_count = sum(1 for item in path.rglob("*") if item.is_file())
        print(f"{path.name}/: {file_count} files")
    else:
        print(f"{path.name}: {path.stat().st_size} bytes")

Deeplearning root: /content/Deeplearning
           dataset                              path  records        experiments
                                                
```

Procedure: Trains a CNN, evaluates it with a confusion matrix, and visualizes learned first-layer filters.

Run the cell and wait for the completion message. The model, numerical results, and plots are written to the corresponding folder under `/content/Deeplearning/outputs`.

Code

```
from pathlib import Path
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import torch
import torch.nn as nn
from sklearn.metrics import ConfusionMatrixDisplay, accuracy_score
from torch.utils.data import DataLoader, TensorDataset


root = Path("/content/Deeplearning") if Path("/content").exists() else Path(__file__).resolve().parents[1]
dataset_dir = root / "datasets"
output_dir = root / "outputs" / "experiment_04"
output_dir.mkdir(parents=True, exist_ok=True)
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
torch.manual_seed(42)

class_names = ["airplane", "automobile", "bird", "cat", "deer", "dog", "frog", "horse", "ship", "truck"]
with np.load(dataset_dir / "cifar10_lab.npz") as dataset:
    X_train = dataset["x_train"].astype("float32") / 255.0
    y_train = dataset["y_train"].astype("int64")
    X_test = dataset["x_test"].astype("float32") / 255.0
    y_test = dataset["y_test"].astype("int64")

X_train = (X_train - 0.5) / 0.5
X_test = (X_test - 0.5) / 0.5
train_tensor = torch.from_numpy(X_train).permute(0, 3, 1, 2)
test_tensor = torch.from_numpy(X_test).permute(0, 3, 1, 2)
train_loader = DataLoader(TensorDataset(train_tensor, torch.from_numpy(y_train)), batch_size=64, shuffle=True)
test_loader = DataLoader(TensorDataset(test_tensor, torch.from_numpy(y_test)), batch_size=256, shuffle=False)


class SimpleCNN(nn.Module):
    def __init__(self):
        super().__init__()
        self.conv1 = nn.Conv2d(3, 16, 3, padding=1)
        self.conv2 = nn.Conv2d(16, 32, 3, padding=1)
        self.pool = nn.MaxPool2d(2, 2)
        self.relu = nn.ReLU()
        self.classifier = nn.Sequential(
            nn.Flatten(),
            nn.Linear(32 * 8 * 8, 128),
            nn.ReLU(),
            nn.Dropout(0.25),
            nn.Linear(128, 10),
        )

    def forward(self, values):
        values = self.pool(self.relu(self.conv1(values)))
        values = self.pool(self.relu(self.conv2(values)))
        return self.classifier(values)


model = SimpleCNN().to(device)
criterion = nn.CrossEntropyLoss()
optimizer = torch.optim.Adam(model.parameters(), lr=0.001)
history_rows = []

for epoch in range(3):
    model.train()
    total_loss = 0.0
    correct = 0
    total = 0
    for images, labels in train_loader:
        images = images.to(device)
        labels = labels.to(device)
        optimizer.zero_grad()
        outputs = model(images)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()
        total_loss += float(loss.item()) * len(images)
        correct += int((outputs.argmax(dim=1) == labels).sum().item())
        total += len(images)
    row = {"epoch": epoch + 1, "loss": total_loss / total, "training_accuracy": correct / total}
    history_rows.append(row)
    print(row)
```

```
model.eval()
all_predictions = []
all_labels = []
with torch.no_grad():
    for images, labels in test_loader:
        outputs = model(images.to(device))
        all_predictions.extend(outputs.argmax(dim=1).cpu().numpy().tolist())
        all_labels.extend(labels.numpy().tolist())

test_accuracy = accuracy_score(all_labels, all_predictions)
history = pd.DataFrame(history_rows)
history["test_accuracy"] = test_accuracy
history.to_csv(output_dir / "cnn_training_history.csv", index=False)
torch.save(model.state_dict(), output_dir / "simple_cnn_cifar10.pt")

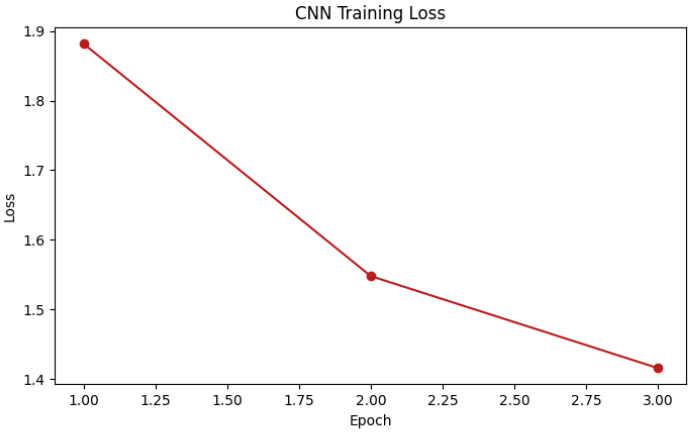
performance_figure, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].plot(history["epoch"], history["loss"], marker="o", color="#b91c1c")
axes[0].set_title("CNN Training Loss")
axes[0].set_xlabel("Epoch")
axes[0].set_ylabel("Loss")
ConfusionMatrixDisplay.from_predictions(
    all_labels,
    all_predictions,
    display_labels=class_names,
    cmap="Blues",
    colorbar=False,
    xticks_rotation=45,
    ax=axes[1],
)
axes[1].set_title(f"CIFAR-10 Confusion Matrix | Accuracy {test_accuracy:.3f}")
performance_figure.tight_layout()
performance_figure.savefig(output_dir / "cnn_evaluation.png", dpi=180, bbox_inches="tight")
plt.show()

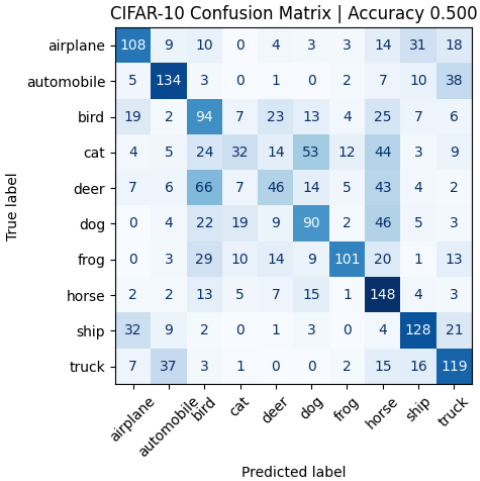
filters = model.conv1.weight.detach().cpu()
filter_figure, filter_axes = plt.subplots(2, 4, figsize=(10, 5))
for index, axis in enumerate(filter_axes.ravel()):
    kernel = filters[index].permute(1, 2, 0).numpy()
    minimum = kernel.min()
    maximum = kernel.max()
    normalized = (kernel - minimum) / (maximum - minimum + 1e-9)
    axis.imshow(normalized)
    axis.set_title(f"Filter {index + 1}")
    axis.axis("off")
filter_figure.suptitle("Learned First-Layer CNN Filters")
filter_figure.tight_layout()
filter_figure.savefig(output_dir / "learned_filters.png", dpi=180, bbox_inches="tight")
plt.show()

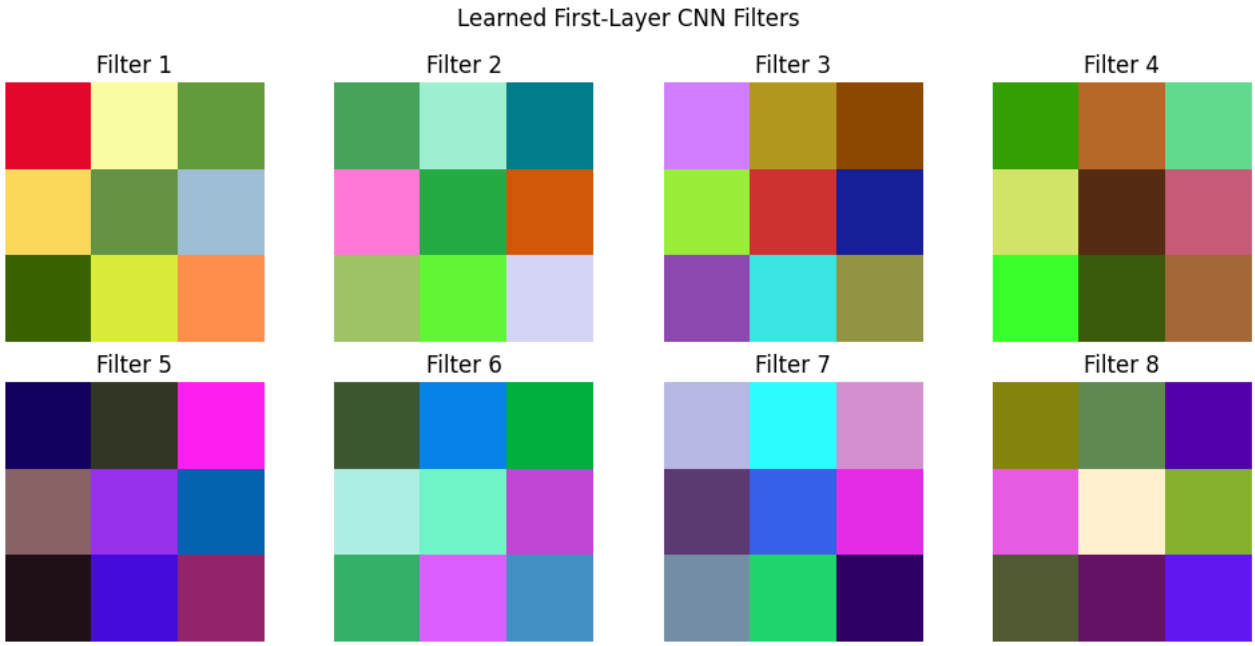
model = model.to("cpu")
if torch.cuda.is_available():
    torch.cuda.empty_cache()

print("Experiment 4 completed")
print(f"CIFAR-10 test accuracy: {test_accuracy:.4f}")
print(f"Output folder: {output_dir}")

{'epoch': 1, 'loss': 1.881644515991211, 'training_accuracy': 0.31675}
{'epoch': 2, 'loss': 1.5478500204086303, 'training_accuracy': 0.4365}
{'epoch': 3, 'loss': 1.415623332977295, 'training_accuracy': 0.487125}
```








```
Experiment 4 completed
CIFAR-10 test accuracy: 0.5000
Output folder: /content/Deeplearning/outputs/experiment_04
```

Output check

Confirm that this experiment saved its results before moving on.

Code

```
from pathlib import Path
output_dir = Path('/content/Deeplearning/outputs/experiment_04')
files = sorted(path for path in output_dir.rglob('*') if path.is_file())
assert files, f'No outputs found in {output_dir}; run the experiment cell first.'
print(f'{len(files)} output files saved in {output_dir}')
for path in files:
    print(path.relative_to(output_dir))
```